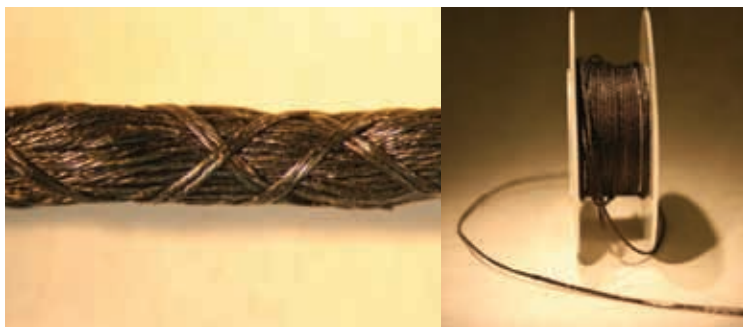




Manufacturing at the nanoscale

water-repellent, anti-reflective, self-cleaning, ultraviolet or infrared-resistant, anti-fog, anti-microbial, scratch-resistant or electrically conductive, among other traits.

What changes at the nanoscale?

While microscale objects (objects with dimensions one-millionth of a metre in size) are widely used, they haven't caused the same excitement as nanoscale materials. The reason is that microscale objects have essentially the same properties as bulk material. However, at the nanoscale the fundamental properties of materials depend on their size, shape and composition in a way that they don't at any other scale. So, the nanoscale is a different kind of small. Size-dependent properties are the reason that nanoscale objects have the potential to significantly impact both, science and industry.



Computer simulation of electron motions within a nanowire

Quantum effects

When particles are created with dimensions of about 1-100 nms, the so-called quantum effects rule the behaviour and properties of particles. Properties such as melting point, fluorescence, electrical conductivity, magnetic permeability and chemical reactivity change as a function of the size of the particle.